Question Paper

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MANIPAL ACADEMY OF HIGHER EDUCATION

Modern Databases for Big Data [BDA 5202]

Marks: 50 **Duration: 90 mins.**

MID-SEMESTER TEST

Answer all the questions.

State any assumptions clearly and precisely.

1)	A group messaging system allows members of a group to exchange text messages. The system does not limit the number of groups that exist in the system. However, the system is required to enforce the following constraints: (C1) A group must have a unique name in the entire system. (C2) A group can have at most 200 members. (C3) A user can create at most 10 groups. (C4) A user can be a member of at most 16 groups. (C5) A group can have one or more admins. (C6) There is no limit on the number of admins for a group. (C7) Each text message can be at most 512 bytes in size. (C8) A response message may refer to a prior message. (C9) There are two types of messages: Simple and Critical. Create a set of syntactically correct definitions for the necessary collections and documents using MongoDB's schema specification language.	(10)
2)	Differentiate between the following concepts: (a) Replica Set and Sharded Cluster in MongoDB architecture. (b) Eventual consistency guarantee versus ACID guarantee.	(10)
3)	Given the group messaging system you described in question (1) above, justify the readConcern and writeConcern you may want to use while performing the following operations: (a) making an existing member the admin of the group. (b) reading messages from a group. (c) sending a critical message to a group. (d) deleting a message for the entire group. (e) deleting a group for which you are an admin.	(10)
4)	For the problem statement in question (1) above, define SQL tables, indices, and constraints that you may require to support a fully functional messaging system.	(10)
5)	Analyze the importance of the following principles and/or ideas: (a) CAP theorem	(10)

(b) BASE transaction model (as against ACID model)